

Fuselage Variables for XFLR5 Panel-Method Optimisation

Variables that can be parameterised and evaluated within an XFLR5-style aerodynamic workflow. The list focuses on fuselage geometry and conditions that can influence the aerodynamic solution, rather than structural or internal engineering variables.

Important: XFLR5's panel method is useful for conceptual aerodynamic comparison, but it is not a high-fidelity fuselage CFD solver. Very complex separated flow, transonic shocks, inlets/outlets and detailed viscous effects should be validated with higher-fidelity CFD later.

1. Primary fuselage geometry

Variable	Primary aerodynamic effect
Fuselage length	Changes wetted area, form/pressure drag and longitudinal aerodynamic characteristics.
Maximum fuselage width	Changes cross-sectional area, wetted area and pressure/form drag.
Maximum fuselage height	Changes cross-sectional area, wetted area and pressure/form drag.
Fineness ratio	Controls slenderness and strongly influences pressure/form drag.
Cross-sectional area distribution	Controls pressure gradients, flow acceleration and pressure drag.
Maximum cross-sectional area	Affects displacement of airflow and pressure drag.
Location of maximum cross-sectional area	Changes pressure recovery and drag distribution.
Nose length	Changes nose pressure gradients and flow separation.
Tail-cone length	Controls aft-body pressure recovery and base drag.
Tail-cone taper/boat-tail angle	Changes pressure recovery and separation at the rear fuselage.
Fuselage taper ratio	Controls longitudinal area change and pressure recovery.

2. Cross-section geometry

Variable	Primary aerodynamic effect
Cross-section shape	Changes pressure distribution, wetted area and separation behaviour.
Width-to-height ratio	Changes cross-flow and pressure distribution around the body.
Cross-sectional aspect ratio	Changes body pressure distribution and interference with nearby surfaces.
Corner radius	Rounded corners generally reduce abrupt separation and pressure losses.
Cross-sectional area at each station	Controls local flow acceleration/deceleration and pressure gradients.
Longitudinal cross-section transitions	Smooth transitions can reduce adverse pressure gradients and drag.

3. Nose geometry

Variable	Primary aerodynamic effect
Nose shape	Controls stagnation pressure, pressure recovery and form drag.
Nose radius	Changes leading-body pressure gradients and local flow behaviour.
Nose curvature	Controls how rapidly the flow accelerates around the nose.
Nose bluntness	Generally increases frontal/pressure drag as bluntness increases.
Nose taper	Controls the longitudinal pressure gradient.

4. Rear fuselage geometry

Variable	Primary aerodynamic effect
Aft-body curvature	Controls pressure recovery and separation.
Base diameter/area	Influences base pressure and base drag.
Boat-tail geometry	Can reduce or increase pressure drag depending on recovery and separation.
Aft-body taper rate	Controls adverse pressure gradient and wake development.

5. Wing-fuselage integration

Variable	Primary aerodynamic effect
Wing longitudinal position on fuselage	Changes wing-body interference and aircraft pitching moment.
Wing vertical position	Changes interference flow and local pressure distribution.
Wing-fuselage intersection geometry	Strongly affects interference drag and local separation.
Wing incidence relative to fuselage	Changes aircraft trim and wing-body flow condition.
Wing-fuselage fairing/fillet geometry	Can reduce interference drag and flow separation.
Wing overlap/penetration geometry	Changes local flow interaction between lifting surface and body.

6. Tail-fuselage integration

Variable	Primary aerodynamic effect
Horizontal-tail attachment location	Changes interference, tail effectiveness and pitching moment.
Vertical-tail attachment location	Changes interference and directional effectiveness.
Tail-fuselage junction geometry	Controls local separation and interference drag.
Tail moment arm	Changes pitch/yaw control authority and stability.

Variable	Primary aerodynamic effect
Tail mounting height relative to fuselage	Changes exposure to fuselage wake and interference.

7. External fuselage aerodynamic features

Variable	Primary aerodynamic effect
Surface roughness	Can increase drag and modify boundary-layer transition.
Local curvature changes	Can create favourable or adverse pressure gradients.
Fairings	Can reduce interference and form drag around junctions.
Fillets	Smooth junction flow and can reduce local separation.
External protrusions	Usually increase local disturbance and parasite drag.
Inlets/outlets represented geometrically	Can alter local pressure and flow if included in the geometry.